Table SI-1. Virtual Screening Results on 60 designed compounds with 3 models, RF-Grover, XGBoost, and MLP-Grover.

Code	SMILES	RF - Grover		XGBoost - Grover		MLP - Grover		Active Hits
		Active Probability	Predicted Label	Active Probability	Predicted Label	Active Probability	Predicted Label	
Ia	O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=C(C=CC=C4)Cl)=O)=CC=C32)C=C1)NO	0.47	Inactive	0.29	Inactive	0.67	Active	1
Ib	O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C32)C=C1)NO	0.44	Inactive	0.32	Inactive	0.67	Active	1
Ic	O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO	0.26	Inactive	0.02	Inactive	0.27	Inactive	0
Id	O=C(CCC1=CC=C(CN2C3=CC(NC(C4=C(C=CC=C4)Cl)=O)=CC=C3N=C2)C=C1)NO	0.47	Inactive	0.25	Inactive	0.67	Active	1
Ie	O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2)C=C1)NO	0.46	Inactive	0.16	Inactive	0.68	Active	1
If	O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO	0.28	Inactive	0.03	Inactive	0.28	Inactive	0

IIa	O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=C(C=CC=C4)Cl)=O)=CC=C32)C=C1)NO	0.29	Inactive	0.16	Inactive	0.21	Inactive	0
IIb	O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC(Cl)=CC=C4)=O)=CC=C32)C=C1)NO	0.37	Inactive	0.33	Inactive	0.61	Active	1
IIc	O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO	0.19	Inactive	0.01	Inactive	0.03	Inactive	0
IId	O=C(CCC1=CC=C(CN2C3=CC(CC(C4=C(C=CC=C4)Cl)=O)=CC=C3N=C2)C=C1)NO	0.31	Inactive	0.16	Inactive	0.22	Inactive	0
IIe	O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2)C=C1)NO	0.37	Inactive	0.33	Inactive	0.62	Active	1
IIf	O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO	0.20	Inactive	0.01	Inactive	0.04	Inactive	0
IIIa	O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO	0.36	Inactive	0.13	Inactive	0.19	Inactive	0
IIIb	O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC(Cl)=CC=	0.44	Inactive	0.28	Inactive	0.46	Inactive	0

	C4)=CC=C32)C=C1)NO							
IIIc	O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.28	Inactive	0.01	Inactive	0.05	Inactive	0
IIId	O=C(CCC1=CC=C(CN2C3=CC(CCC4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.36	Inactive	0.13	Inactive	0.19	Inactive	0
IIIe	O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO	0.43	Inactive	0.31	Inactive	0.50	Inactive	0
IIIf	O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO	0.27	Inactive	0.03	Inactive	0.06	Inactive	0
IVa	O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO	0.48	Inactive	0.40	Inactive	0.75	Active	1
IVb	O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO	0.55	Active	0.86	Active	0.91	Active	3
IVc	O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.33	Inactive	0.01	Inactive	0.11	Inactive	0

IVd	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.48	Inactive	0.24	Inactive	0.75	Active	1
IVe	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO	0.57	Active	0.64	Active	0.90	Active	3
IVf	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO	0.36	Inactive	0.03	Inactive	0.12	Inactive	0
Va	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.35	Inactive	0.25	Inactive	0.67	Active	1
Vb	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.40	Inactive	0.15	Inactive	0.85	Active	1
Vc	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.27	Inactive	0.01	Inactive	0.09	Inactive	0
Vd	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.36	Inactive	0.22	Inactive	0.69	Active	1
Ve	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4	0.41	Inactive	0.19	Inactive	0.86	Active	1

	)=CC=C3N=C2)C=C1)=O)NO							
Vf	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.27	Inactive	0.04	Inactive	0.10	Inactive	0
VIa	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.62	Active	0.88	Active	0.42	Inactive	2
VIb	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.68	Active	0.88	Active	0.70	Active	3
VIc	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.42	Inactive	0.19	Inactive	0.03	Inactive	0
VId	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.63	Active	0.69	Active	0.42	Inactive	2
VIe	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)=O)NO	0.68	Active	0.88	Active	0.69	Active	3
VIf	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.41	Inactive	0.17	Inactive	0.03	Inactive	0

VIIa	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.48	Inactive	0.17	Inactive	0.11	Inactive	0
VIIb	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.49	Inactive	0.20	Inactive	0.28	Inactive	0
VIIc	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.32	Inactive	0.01	Inactive	0.01	Inactive	0
VIId	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.48	Inactive	0.17	Inactive	0.10	Inactive	0
VIIe	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)=O)NO	0.51	Active	0.20	Inactive	0.26	Inactive	1
VIIf	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.31	Inactive	0.02	Inactive	0.01	Inactive	0
VIIIa	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO	0.52	Active	0.60	Active	0.53	Active	3
VIIIb	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(Cl)=	0.52	Active	0.45	Inactive	0.78	Active	2

	CC=C4)=CC=C32)C=C1)NO							
VIIIc	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.35	Inactive	0.01	Inactive	0.05	Inactive	0
VIIId	O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.50	Inactive	0.41	Inactive	0.53	Active	1
VIIIe	O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO	0.52	Active	0.45	Inactive	0.77	Active	2
VIIIf	O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO	0.33	Inactive	0.01	Inactive	0.05	Inactive	0
IXa	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=C(Cl)C=CC=C4)=O)=CC=C32	0.60	Active	0.94	Active	0.96	Active	3
IXb	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C32	0.57	Active	0.94	Active	0.96	Active	3
IXc	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C32	0.52	Active	0.31	Inactive	0.88	Active	2

IXd	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=C(Cl)C=CC=C4)=O)=CC=C3N=C2	0.62	Active	0.95	Active	0.97	Active	3
IXe	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2	0.56	Active	0.93	Active	0.97	Active	3
IXf	O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C3N=C2	0.51	Active	0.44	Inactive	0.89	Active	2
Xa	O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO	0.58	Active	0.73	Active	0.78	Active	3
Xb	O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.39	Inactive	0.09	Inactive	0.14	Inactive	0
Xc	O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.60	Active	0.73	Active	0.78	Active	3
Xd	O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO	0.65	Active	0.94	Active	0.91	Active	3
Xe	O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)	0.40	Inactive	0.08	Inactive	0.14	Inactive	0

	=CC=C3N=C2)C=C1)NO							
Xf	O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO	0.64	Active	0.95	Active	0.92	Active	3